

Do Topic-Level Budgets and Intra-Group Credit Redistribution Compose?

A Pre-Registered Study of Pass@ k Shrinkage in Small Language Models

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DRAFT. Red numbers are pre-registered expectations, not results; violet text depends on the results; blue marks open items. Edit only as described in EDIT_GUIDE.md.

Abstract—Reinforcement learning with verifiable rewards (RLVR) raises the single-sample accuracy of language models but often lowers Pass@ k at large k , shrinking the set of problems a model can solve at all. Recent mitigations act at two different granularities: *within* a rollout group, by redistributing credit away from over-represented solution modes, or *across* prompts, by reweighting problems or tasks. Whether the two address distinct components of shrinkage has not been tested. We report a pre-registered 2×3 factorial study on Qwen2.5-0.5B-Instruct trained with GRPO on MATH, crossing a macro factor (none vs. a budget on positive advantage mass across the seven MATH subjects) with a micro factor (none, the per-prompt SELF filter, or mean-preserving rarity redistribution over solution modes), with 3 seeds per cell. On MATH-500, vanilla GRPO raises Pass@1 from 30.0% to 36.5% while Pass@32 falls from 71.5% to 66.5%. The macro-micro composite shrinks least, but neither pre-registered primary contrast survives Holm correction, and the macro-micro interaction is bounded within $[-0.27, +0.63]$ pp per unit $\ln k$.^[R1] Mode-entropy and boundary entry/exit diagnostics, logged for every condition, show where each level acts. The contribution is not a new gate but a pre-registered complementarity test and a reusable diagnostic protocol at a compute scale (one 24GB GPU) where a bounded null is as informative as a gain.

Index Terms—reinforcement learning with verifiable rewards, GRPO, Pass@ k , reasoning boundary, small language models, credit assignment, pre-registration

I. INTRODUCTION

Group Relative Policy Optimization (GRPO) [1] and related RLVR methods have become the standard way to improve mathematical reasoning in open language models. A consistent finding, however, is that the gains concentrate at small sampling budgets: the trained model wins at Pass@1, while the base model solves more problems when many samples are drawn [2]. This Pass@ k inversion means RLVR can narrow the reasoning boundary instead of extending it [3], [4], and recent work traces it to the concentration of credit: within a group of rollouts, whichever correct solution mode is sampled

most often receives most of the positive gradient, and the policy collapses onto it [3], [5]. The problem is sharper for small language models (SLMs), whose usefulness often rests on sampling-based coverage (majority voting, verifier reranking) rather than on a single greedy answer.

Two families of mitigations have emerged. *Intra-group* methods redistribute credit among the rollouts of one prompt: rare correct modes receive more credit than common ones [5]–[10]. *Cross-prompt* methods reweight whole problems or tasks, by pass rate [11], success count [12], task-level robustness [13], or by training only on prompts whose greedy answer still fails (SELF [3]). The two act on different axes: one on how credit is shared among solutions of a single problem, the other on how the gradient budget is spread over the problem distribution. If shrinkage has components along both axes, combining the levels should help beyond either alone; if it is driven by one mechanism, they are redundant. To our knowledge this has not been tested: ReCo combines two complementary corrections, but both operate within a single prompt [6].

We test it directly, with a design fixed before any main run. Our contributions are:

- A pre-registered 2×3 factorial on Qwen2.5-0.5B-Instruct [14] trained on MATH [15]: a *macro* factor (none vs. a topic-level budget on positive advantage mass, CB) crossed with a *micro* factor (none, the per-prompt SELF filter, or intra-group mode-rarity redistribution, MEG), with seed-paired primary contrasts and Holm correction.
- CB, a stateful topic-level gate that measures each topic’s share of positive advantage mass and throttles overspending topics, composed multiplicatively with any per-rollout micro gate.
- A measurement protocol for SLM-scale RLVR: a solution-mode partitioner chosen by a pre-registered calibration rule with a random-partition control, mode entropy and boundary entry/exit counts for every condition, and rollouts-per-problem reporting [12].

- Evidence that, at 0.5B and this compute budget, the two levels are largely redundant: their interaction is bounded near zero, and the diagnostics locate the effect in the micro term.^[R2]

Code, logs and the pre-registration are released at [\[TODO: repository URL\]](#).

II. RELATED WORK

Pass@ k shrinkage under RLVR. Yue et al. [2] showed that RLVR-trained models are overtaken by their base models at large k . Nguyen et al. [3] attribute this to negative interference and winner-take-all reinforcement, and propose SELF, which trains only on problems whose greedy answer is wrong; Zhou [4] diagnoses the inversion further. Yuan et al. [12] view diversity collapse as overtraining, argue that aggregate Pass@ k declines partly reflect few rollouts per problem, and gate problems by success count (BBG). Other responses change the optimizer or the search: evolution strategies [16], tree-structured exploration [17], and a decomposition of policy-gradient weights [18]. At SLM scale, GRPO training dynamics [19], difficulty scaling [20] and reward granularity [21] have been studied, but not shrinkage mitigation.

Intra-group credit redistribution. Cue-GRPO [5] redistributes positive credit over a partition of the correct rollouts by rarity, leaving incorrect rollouts untouched and restoring the mean; we adopt its rule as our micro term and say so explicitly. ReCo [6] reweights against distributional concentration at the response and token level; GCPO [7], DRA-GRPO [8], EDAS [9] and TreeAdv [10] reshape advantages using diversity or structure within the group. Lee et al. [22] show that embedding similarity tracks phrasing more than strategy, that word-bigram overlap tracks approach better, and that only 0.7% of GSM8K problems admit more than one approach against 33% on MATH; this motivates both our training set and our calibrated partitioner.

Cross-prompt reweighting. MT-GRPO [13] reweights explicit tasks for worst-task reliability; CurveRL [11] reweights prompts by their pass-rate distribution; BBG gates problems by success count [12]; SELF selects by greedy failure [3]. CB differs in granularity (latent topics rather than tasks or single prompts), in signal (each topic’s share of positive advantage mass), and in mechanism (a stateful multiplicative gate). Our question is not which gate is best but whether a topic-level and an intra-group intervention compose.

III. METHOD

A. GRPO with gated advantages

For a prompt x the policy π_θ samples a group of G completions $y_1, \dots, y_G$ with binary rewards $r_i \in \{0, 1\}$ (answer correctness). GRPO [1] uses the group-normalized advantage

$$\hat{A}_i = \frac{r_i - \text{mean}(r)}{\text{std}(r) + \epsilon}. \quad (1)$$

Every condition in this study multiplies it by a non-negative gate weight, $\tilde{A}_i = g_i \hat{A}_i$ with $g_i = g_c^{\text{mac}} g_i^{\text{mic}}$, and minimizes

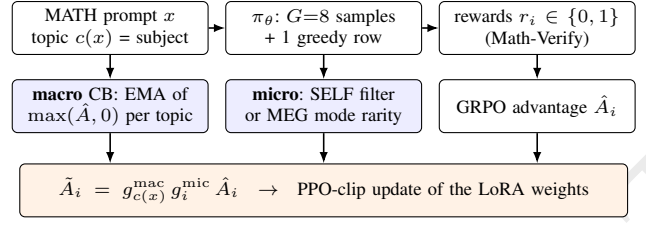


Fig. 1. Gated GRPO. The macro gate sees which topic a prompt belongs to and how much positive advantage that topic has absorbed in past steps; the micro gate sees the rollouts (and greedy answer) of the current group. Both scale the same advantage.

TABLE I
THE 2×3 FACTORIAL. COMPOSITES MULTIPLY THE MACRO AND MICRO GATES.

macro factor	micro factor		
	none	SELF filter	MEG redistribution
none	VANILLA	SELF	MEG
CB topic budget	CB	CB $\times$ SELF	CB $\times$ MEG (HEG-GRPO)

the PPO-clip loss [23], averaged over all T completion tokens in the batch [24]:

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_i \sum_{t=1}^{|y_i|} \min(\rho_{i,t} \tilde{A}_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) \tilde{A}_i), \quad (2)$$

where $\rho_{i,t}$ is the token probability ratio. Updates are on-policy and use no KL penalty. Vanilla GRPO is $g_i \equiv 1$. Fig. 1 summarizes the pipeline.

B. Macro gate: topic budget (CB)

Topics are the $C = 7$ MATH subjects. After each step, CB updates an exponential moving average of each topic’s *spend*, the mean positive advantage mass of its rollouts in that step, $s_c \leftarrow (1 - \eta) s_c + \eta \text{mean}_{i:c(x_i)=c} \max(\hat{A}_i, 0)$, with $\eta = 0.05$. Spend counts only positive mass, so corrective negative gradient never uses up a topic’s budget. With the overspend ratio $\rho_c = s_c / \frac{1}{C} \sum_{c'} s_{c'}$, the gate is

$$g_c^{\text{mac}} = \delta^{\max(0, \rho_c - \vartheta)}, \quad \vartheta = 1.5, \delta = 0.98, \quad (3)$$

so topics that absorb more than ϑ times the average positive mass are throttled. Since $\rho_c \leq C$, $g_c^{\text{mac}} \in [\delta^{C-\vartheta}, 1]$. **[TODO: CB strength: with $C=7$ the gate never drops below 0.90; decide ϑ, δ from the pilot (EDIT_GUIDE §8)]**

C. Micro gates: SELF filter and mode-rarity redistribution (MEG)

SELF. Following [3], a prompt whose greedy answer is already correct gets $g_x^{\text{self}} = \lambda$ ($\lambda = 0$, a hard filter), otherwise 1. The greedy answer is decoded as one extra row of the same generation call (its logits are pinned to the argmax), so it costs one sequence per prompt. We implement the selection rule only, not SELF’s forward-KL term.

TABLE II
PARTITIONER CALIBRATION ON BASE-MODEL ROLLOUTS (GROUPS WITH ≥ 4 CORRECT). RULE: MEDIAN MODES IN [2, 4], CLOSEST TO 3; * = SELECTED.

MiniLM cosine τ	0.80	0.85	0.90	0.95	
median modes	1.0	1.0	1.5	2.5	
multi-mode groups (%)	5	10	35	70	
Word-bigram threshold	0.30	0.40	0.50	0.60	0.70
median modes	3.5	3.0*	2.0	1.5	1.0
multi-mode groups (%)	92	78	55	35	15

MEG. Let $\mathcal{C}$ be the N correct rollouts of a group, partitioned into solution modes, and let $|M(i)|$ be the size of the mode containing rollout i . Following Cue-GRPO [5],

$$w_i = \frac{N |M(i)|^{-\alpha}}{\sum_{j \in \mathcal{C}} |M(j)|^{-\alpha}} \quad (i \in \mathcal{C}), \quad w_i = 1 \quad (i \notin \mathcal{C}), \quad (4)$$

with $\alpha = 0.8$; weights are clipped to $[0.3, 3]$ and rescaled to mean 1 over $\mathcal{C}$. Because the mean is preserved, MEG changes *which* correct solutions receive credit, not how much; an earlier down-weight-only version was confounded with a lower learning rate. We call this term the mode-equalizing gate.

Mode partitioner. Modes come from agglomerative clustering of the correct completions, either on MiniLM [25] sentence embeddings [26] (cosine threshold) or on word-bigram Jaccard distance. Because embeddings track phrasing more than strategy [22], the partitioner is not hand-tuned: a pre-registered rule, applied to base-model rollouts before training, picks the candidate whose median number of modes per group lies in [2, 4] and is closest to 3 (ties go to bigram). It selected **word-bigram Jaccard, threshold 0.4** (median 3.0 modes; **75%** of groups with more than one mode; Table II). The rejected embedding candidates illustrate the risk: at the thresholds commonly used, **they place almost every group's correct rollouts in a single mode**^[R15], which would turn MEG into a no-op.

A control condition reuses the composite gate with random mode labels (same number of modes per group).

D. Factorial design

Table I lists the six conditions. Composites multiply the two gates. Two exploratory extras complete the picture: CB $\times$ MEG with random partitions, and a GRPO-compatible version of BBG's success-count gate [12] (without its signed REINFORCE term for all-wrong groups, which group-normalized GRPO cannot express). Algorithm 1 gives one update for all conditions.

IV. EXPERIMENTAL SETUP

Training. We train Qwen2.5-0.5B-Instruct [14] with LoRA [27] adapters on the MATH training split [15] (7,496 problems with a boxed answer), prompted with the standard “reason step by step and put the final answer in `\boxed{\}`” instruction; Table III lists every setting. Rewards are binary correctness checked with Math-Verify [28] under a time limit,

Algorithm 1 One update of gated GRPO (all conditions)

Require: policy π_θ (LoRA), topic map $c(\cdot)$, CB state $s \in \mathbb{R}^C$, gate configuration
1: draw 16 prompts; per prompt, sample $G=8$ rollouts plus one greedy row in one generation call
2: $r_i \leftarrow$ Math-Verify correctness; $\hat{A}_i \leftarrow$ group-normalized advantage (1)
3: $g_i^{\text{mac}} \leftarrow \delta^{\max(0, \rho_c(x) - \vartheta)}$ if CB is on, else 1 $\triangleright$ (3)
4: **if** micro factor is SELF **then**
5: $g_i^{\text{mic}} \leftarrow \lambda$ if the greedy answer to x is correct, else 1
6: **else if** micro factor is MEG **then**
7: partition the correct rollouts into modes (random labels in the control); $g_i^{\text{mic}} \leftarrow w_i$ (4)
8: **else**
9: $g_i^{\text{mic}} \leftarrow 1$
10: **end if**
11: $\tilde{A}_i \leftarrow g_i^{\text{mac}} g_i^{\text{mic}} \hat{A}_i$; one AdamW step on (2)
12: update s_c with this step's positive advantage mass per topic
13: log mode entropy, mean positive weight, SELF selection rate (every condition)

TABLE III
TRAINING AND EVALUATION SETTINGS, IDENTICAL FOR ALL CONDITIONS.

Model	Qwen2.5-0.5B-Instruct, bf16 weights
Adapters	LoRA $r=16$, $\alpha=32$, dropout 0, all projections
Optimizer	AdamW, lr 1×10^{-5} , cosine, 10% warm-up, grad-norm clip 1.0
Batch	16 prompts $\times G=8$ rollouts = 128 rollouts per step
Steps	400 ; MATH levels 1-5
Sampling	$T=0.7$, top- p 0.95, no top- k , $\leq 1,024$ new tokens
Loss	PPO-clip $\varepsilon=0.2$, on-policy, token-level mean, no KL
CB / SELF / MEG	$\eta=0.05$, $\vartheta=1.5$, $\delta=0.98$ / $\lambda=0$ / $\alpha=0.8$, clip $[0.3, 3]$
Evaluation	$n=32$ samples, $T=0.6$, top- p 0.95; MATH-500 (1,024 tokens), GSM8K 500-problem subset (512 tokens)
Seeds	3 per condition
Hardware	one NVIDIA A10G 24GB (AWS g5.xlarge)

with a guarded symbolic fallback; a strict tag-format reward was dropped after it proved identically zero, hence inert under group normalization. The budget amounts to **0.85** epochs, or **6.8** rollouts per training problem. Seed s fixes the data order and LoRA initialization across conditions, so conditions are compared in seed-paired blocks.

Evaluation. Every model, including the base model, is scored on the same problems: all of MATH-500 [29] (in-distribution, primary; 1,024 tokens) and a fixed 500-problem subset of the GSM8K test set [30] (out-of-distribution; 512 tokens), with $n = 32$ samples per problem at temperature 0.6 and top- p 0.95. Pass@ k for $k \in \{1, 2, 4, 8, 16, 32\}$ uses the unbiased estimator [31] $1 - \binom{n-c}{k} / \binom{n}{k}$ for c correct samples. MATH-500 shares no problem statement with the training split; 7 of its problems (1.4%) are template siblings of training problems with a different question or numbers.

Outcome. The pre-registered primary outcome is the *shrinkage slope*: the least-squares slope of $\Delta(k) = \text{Pass}@k(\text{run}) - \text{Pass}@k(\text{base})$ against $\ln k$, reported in percentage points per unit $\ln k$. A negative slope means gains at small k turn into losses at large k ; higher means less shrinkage. We also count *boundary exits* (problems the base model solves at least once in 32 samples and the trained model never does) and *entries*

TABLE IV

MATH-500 PASS@ k (%), MEAN OVER SEEDS), SHRINKAGE SLOPE (PP PER UNIT $\ln k$; HIGHER IS BETTER) AND BOUNDARY ENTRIES/EXITS (OF 500 PROBLEMS). ROWS BELOW THE RULE ARE EXPLORATORY EXTRAS.

Condition	P@1	P@4	P@16	P@32	Slope	In / Out
Base model	30.0	49.8	65.5	71.5	—	—
VANILLA	36.5	51.0	62.0	66.5	-3.41	8.3 / 24.7
SELF	35.4	50.9	62.8	67.6	-2.76	9.3 / 20.7
MEG	35.8	51.8	64.0	68.9	-2.50	10.0 / 18.3
CB	36.2	51.0	62.4	67.0	-3.17	8.7 / 22.3
CB×SELF	35.2	51.1	63.2	68.1	-2.56	9.0 / 19.7
CB×MEG	35.6	52.4	65.1	70.1	-2.09	10.7 / 15.3
CB×MEG, rand. modes	36.1	51.1	62.5	67.2	-3.08	8.7 / 22.0
BBG-style gate	35.0	51.1	63.4	68.4	-2.41	9.7 / 19.0

(the reverse).

Pre-registered analysis. All contrasts are linear combinations of condition slopes, paired by seed. The two primary contrasts are P1, CB×MEG − CB×SELF (intra-group redistribution vs. per-prompt filtering, both under the budget), and P2, CB×MEG − MEG (does the macro level add to the micro level?). Secondary contrasts are the two macro×micro interactions (S1 for MEG, S2 for SELF), reported as 95% bounds rather than verdicts, and the macro main effect (S3); E1–E3 are exploratory. Each contrast gets a t -interval over seeds and a hierarchical bootstrap that resamples seeds and problems jointly (2,000 resamples). An effect is claimed only if its Holm-adjusted [32] bootstrap $p < 0.05$ and its bootstrap interval excludes zero. Diagnostics, logged for every condition: normalized mode entropy of the correct rollouts in each group, the mean gate weight on positive-advantage rollouts, and the share of prompts SELF selects. The pre-registration also fixes how outcomes are worded: a significant P1 or P2 means the composition helps beyond its parts; if neither is significant and the intervals are narrow, the result is a bounded null; if vanilla GRPO shows no shrinkage, that is itself reported and the contrasts are still given.

Reproducibility. The pre-registration (committed [TODO: date + commit hash] before any main run), configs, per-step logs, sampled rollout groups for auditing the partitions, evaluation correctness matrices, and the script that generates every number in this paper are released at [TODO: repository URL].

V. RESULTS

A. Shrinkage at 0.5B

Vanilla GRPO reproduces the inversion at this scale^[R3]: Pass@1 rises from 30.0% to 36.5%, Pass@32 falls from 71.5% to 66.5% (Table IV), and the curves cross near $k \approx 6$ (Fig. 2, left). 24.7 problems leave the boundary while 8.3 enter it. Because aggregate shrinkage depends on how often each training problem is visited [12], we note that each training problem received 6.8 rollouts on average; the effect is therefore measured in the low-visit regime typical of SLM budgets.

B. Pre-registered contrasts

TABLE V

CONTRASTS ON THE MATH-500 SHRINKAGE SLOPE (PP PER UNIT $\ln k$), PAIRED BY SEED. CI: HIERARCHICAL BOOTSTRAP OVER SEEDS AND PROBLEMS. p : HOLM-ADJUSTED FOR P1–P2, UNADJUSTED OTHERWISE. GSM8K: ESTIMATE OF THE SAME CONTRAST OUT OF DISTRIBUTION.

Contrast	Est.	95% CI	p	GSM8K
P1 CB×MEG − CB×SELF	+0.47	[+0.02, +0.92]	0.085	+0.09
P2 CB×MEG − MEG	+0.41	[−0.04, +0.86]	0.085	+0.12
S1 CB×MEG interaction	+0.18	[−0.27, +0.63]	0.443	+0.03
S2 CB×SELF interaction	−0.03	[−0.48, +0.42]	0.901	+0.00
S3 CB main effect	+0.28	[−0.17, +0.73]	0.218	+0.10
E1 CB×MEG − Vanilla	+1.31	[+0.86, +1.76]	<0.001	+0.50
E2 CB×MEG − random modes	+0.99	[+0.54, +1.44]	<0.001	+0.38
E3 CB×MEG − BBG-style	+0.32	[−0.13, +0.77]	0.164	−0.00

Every gate has a higher slope than vanilla (Table IV), and CB×MEG shrinks least^[R4] (−2.09 vs. −3.41). Neither primary contrast clears the pre-registered bar, however (Table V): P1 = +0.47 (CI [+0.02, +0.92], Holm $p = 0.085$) and P2 = +0.41 (CI [−0.04, +0.86], Holm $p = 0.085$) both point toward the composite, but their intervals include zero.^[R5] The CB×MEG interaction lies within [−0.27, +0.63] and the CB×SELF interaction within [−0.48, +0.42], so any super-additive benefit of composing the levels is at most a fraction of vanilla’s shrinkage slope; the macro main effect (S3 = +0.28) is small.^[R6] Following the pre-registration we report these as bounds, not as evidence of absence.

C. What each level does

Mode entropy starts at 0.46 (the same base model for every condition) and decays under vanilla GRPO, while both MEG conditions retain it^[R7] (Fig. 2, right; training means 0.31 for VANILLA vs. 0.43 for CB×MEG). The macro gate, in contrast, rarely binds^[R8]: the mean weight on positive-advantage rollouts is 0.98 under CB, against 0.55 under SELF, which filters 58% of prompts on average. Shrinkage is also uneven across the topics the macro gate acts on (Fig. 3): under vanilla GRPO, Pass@32 changes by −8.2 pp in Geometry but −2.2 pp in Prealgebra. CB flattens this profile without reducing the average loss, whereas both MEG conditions reduce it in every subject.^[R16]

Boundary exits fall^[R9] from 24.7 (VANILLA) to 15.3 (CB×MEG). Replacing the calibrated modes with random labels removes most of the gain, so the effect depends on the partition rather than on reweighting noise^[R10] (E2 = +0.99, CI [+0.54, +1.44]); the BBG-style gate (E3 = +0.32) lies between vanilla and the MEG conditions^[R11].

Out of distribution. On GSM8K the same ordering holds with smaller magnitudes^[R12]: slopes are −1.58 for VANILLA and −1.09 for CB×MEG (P1 = +0.09, CI [−0.36, +0.54]). Training on MATH thus does not trade in-distribution coverage for out-of-distribution coverage.^[R12]

VI. DISCUSSION AND LIMITATIONS

Read together, the results support a simple account: at this scale the shrinkage that GRPO induces is concentrated

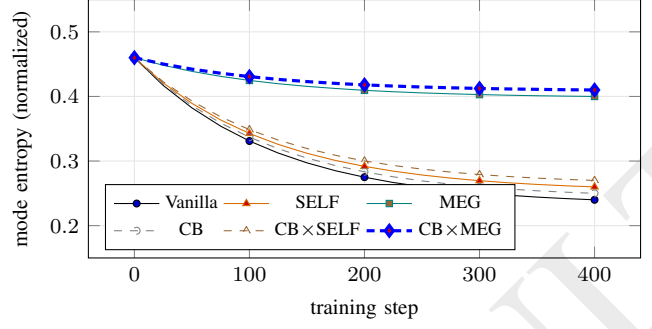
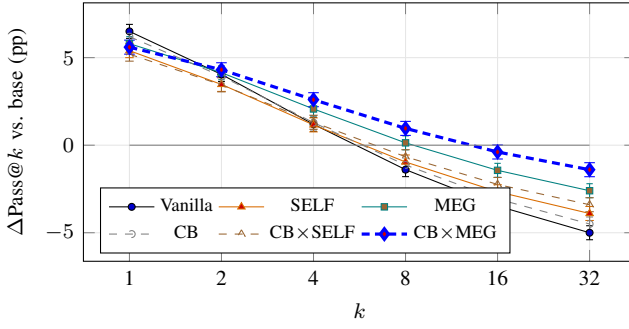


Fig. 2. Left: change in MATH-500 Pass@ k relative to the base model (mean $\pm$ s.d. over seeds); a curve that falls below zero at large k has shrunk the reasoning boundary. Right: normalized entropy of the solution-mode distribution among correct rollouts during training (mean over seeds).

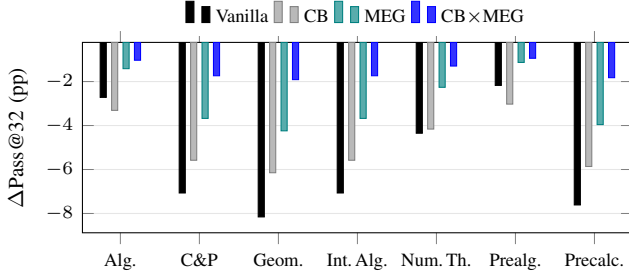


Fig. 3. Change in MATH-500 Pass@32 relative to the base model by subject, i.e. by the macro gate’s topics (mean over seeds).

within groups, where one correct mode crowds out others, and an intervention at that level captures most of what can be recovered; a topic-level budget adds little because topics are not where the gradient concentrates.^[R13] This is a statement about 0.5B models in a low-visit regime, and the bounds in Table V are what a larger study would need to beat.

Limitations. (i) One model family at one size; a 1.5B replication was planned as optional [TODO: state whether it was run]. (ii) Three seeds bound the power for interactions, which is why they are reported as bounds. (iii) Solution modes are proxies: bigram and embedding partitions approximate strategy, and the random-partition control only shows that the partition matters, not that it equals strategy. (iv) SELF is implemented without its forward-KL term, and BBG without its signed loss for all-wrong groups. (v) LoRA rather than full fine-tuning, a single RL algorithm, and short training (6.8 rollouts per problem). (vi) The macro gate’s strength is fixed by the pre-registration; a stronger budget might behave differently.

VII. CONCLUSION

We asked whether a topic-level gradient budget and intra-group credit redistribution address distinct components of Pass@ k shrinkage in a small model. In a pre-registered 2×3 factorial on Qwen2.5-0.5B-Instruct, the composite shrinks least, but the benefit of composing the two levels is bounded near zero, and the diagnostics locate the effect in the intra-group term^[R14]. The protocol (calibrated partitions with a

random control, diagnostics for every condition, seed-paired contrasts with Holm correction) runs on a single 24 GB GPU in 165 GPU-hours, and we release it so that larger studies can test the same question at scale.

ACKNOWLEDGMENT

[TODO: AWS credits (hackathon name), supervisor, compute]. Total compute: 165 A10G GPU-hours (96 for training), about \$166.

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